

International Journal of Sustainable Transportation

ISSN: 1556-8318 (Print) 1556-8334 (Online) Journal homepage: www.tandfonline.com/journals/ujst20

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To cite this article: Paolo Beria (2016) Effectiveness and monetary impact of Milan's road charge, one year after implementation, International Journal of Sustainable Transportation, 10:7, 657-669, DOI: 10.1080/15568318.2015.1083638

To link to this article: https://doi.org/10.1080/15568318.2015.1083638

Published online: 31 May 2016.

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INTERNATIONAL JOURNAL OF SUSTAINABLE TRANSPORTATION
2016, VOL. 10, NO. 7, 657–669
http://dx.doi.org/10.1080/15568318.2015.1083638

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Effectiveness and monetary impact of Milan's road charge, one year after implementation

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ABSTRACT

In early 2008, a pollution charge was introduced in the central Area of Milan. Although initially it positively affected traffic volume, after some years its effect wore off due to the replacement of older vehicles with newer and exempt vehicles. As effectiveness decreased, in June 2011 the Milan citizens voted in a referendum on whether to extend the charge to all vehicles. The result was clearly in favor of the redesign and extension of the toll. In January 2012 the new policy was put in force under the name of "Area C." The new pricing uses the same technological infrastructure and encompasses the same Area but evolves into congestion pricing aimed at reducing traffic in the compact and crowded central city. The present article discusses the history of the Milan road-pricing case and presents some quantitative analyses of the overall effectiveness of the measure, based on official data. In addition, we calculate, using cordon data, the distribution of entrances per vehicle category and the related yearly cost paid by users. The analyses show the measure's effectiveness, in line with expectations, and calculate the economic impact of the pricing on different user groups. Such impact appears quite limited, as the vast majority of users cross the cordon occasionally, and the most affected user groups (e.g., residents) are granted with more favorable conditions.

ARTICLE HISTORY
Received 19 May 2014
Revised 23 March 2015
Accepted 12 August 2015

KEYWORDS
Milan; road charge; transport

1. Introduction

On 2 January 2008, the municipality of Milan made a first attempt to introduce urban road pricing through a pollution charge called Ecopass. The Ecopass Area covered approximately 8 km^2 in the central sector of the city, delimited by Milan's 16th-century walls. The Area was actually very small compared to the city's total surface Area (181 km^2) and even more so when compared to the entire metropolitan Area. The Ecopass trial period lasted until 31 December 2011, when, before deciding whether to extend it or to terminate it, the city mayor of the time announced a referendum on the topic. Given the positive result of the referendum, in 2012 the pricing was upgraded by the new mayor to a more conventional congestion charge, named Area C, and applied to the very same zone. Area C's scope was similar to road charges in London (started in 2003) and Stockholm (implemented in 2006)—that is, to internalize congestion costs and only eventually to reduce pollution.

In this article we outline the history and the principles at the root of the Milan road-charge policy and describe its main effects and impacts. Elaborating on publicly available data, our objective is to quantitatively discuss the most relevant facts related to the charge one year after its implementation, both in terms of effectiveness (traffic and pollution reduction, revenues) and of current distribution of impacts. In particular, we aim to shed light on which users feel the most impact of the policy by analyzing how five user groups (residents, vehicles associated to commercial activities, other cars, low-emission exempt vehicles, authorized vehicles) are actually affected by the charge in terms of frequency of entrances and fees paid. The results of this analysis can help explain the good acceptability rate of the charge, which was well received by city users, and give insight on the transferability of the policy to other cities. The limit of the approach lay in the impossibility of typifying the users' profiles in a more detailed way than using just the five groups, because single-user observations are not yet available. Similarly, we do not know how groups' single individuals behaved before the pricing was introduced. Consequently, we can quantify the difference among ex ante and ex post only in aggregate terms (number of entrances).

The article is structured as follows: Section 2 contextualizes Milan within the main European experiences of urban road charges. Section 3 tells the history of the policy, from 2007 to 2013, and outlines its characteristics. Section 4 presents the effects of the policy, comparing 2011 (with the pollution charge) to 2012 (when the new congestion charge became operative). Section 5 discusses how the different groups of users crossed the cordon during priced hours and to what degree the pricing economically impacted them. Section 6 concludes, drawing useful lessons for other cities.

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2. Urban road charge experiences

The first application of the apparently simple idea of road pricing[1] to control congestion took place in Singapore in June 1975. It proved to be effective at reducing congestion by increasing speed and shifting traffic to less-congested roads, to off-peak periods, and to public transport, which doubled in share (Phang & Toh, 2004).

Following Singapore's path, three Norwegian cities—Bergen (1986), Oslo (1990), and Trondheim (1991)—also adopted road-pricing schemes. The results in terms of traffic reduction were negligible (Lian, 2008), but the declared goal of the policy was mainly to raise funds to invest in transportation infrastructure, especially roads (Ieromonachou, Potter, & Warren, 2006). In general, the population did not welcome the charge, despite it being reinvested (Odeck & Bråthen, 1997; Bråthen & Odeck, 2009), and considered it to be an extra tax, not a solution to congestion problems. In 2001 Stavanger also introduced a toll, this time aimed at controlling congestion.

The best-known case of urban road pricing was adopted by London in 2003. The effects (Leape, 2006) of the charging were significant. Private car trips in central London reduced by 33% during peak hours (65,000–70,000 car trips avoided). Congestion, in terms of minutes of delay, dropped an average of 30% in one year, and traffic around the cordon increased between 2% and 6%. On the other hand, the resulting running costs rose very high, draining up to two-thirds of revenues. For this reason, the financial break-even point depended on fines (Mackie, 2005).[2] In addition, a positive impact on pollution is questioned (Atkinson et al. 2009).

More recently, in 2006, Stockholm introduced a toll system as part of a broader policy package. While earlier road charges had been limited to a small Area, the Stockholm cordon included the majority (two-thirds) of city inhabitants. Revenues were earmarked for road investments. The city of Stockholm used the term *environmental charge*, but its effects (Eliasson, Hultkrantz, Nerhagen, & Smidfelt Rosqvist, 2009; Börjesson, Eliasson, Hugosson, & Brundell-Freij, 2012) were more closely related to traffic. Travel times decreased due to less congestion, public-transport modal share increased, and car modal share decreased, even if only part of the effect was directly related to the charge (Kottenhoff & Brundell-Freij, 2009). The system cost less than London's but was still significant, draining 25% of the total revenues (May et al. 2010). Eventually, Gothenburg followed Stockholm and implemented congestion pricing in January 2013, also aimed at financing an investment package (Börjesson, Eliasson, & Hamilton, 2014). However, the trial was recently rejected by a referendum in 2014.

In 2007 Valletta, the capital city of Malta, introduced a charge to enter in the very center of the city to control increasing congestion and the inefficient consumption of parking space (Attard & Ison, 2010; Attard & Enoch, 2011).

The mentioned cities are not alone in considering the introduction of a road-pricing scheme. Others (such as New York, San Francisco, West Midlands, and some other European cities) experienced political debates or public consultations. A few organized referenda to ask the citizens their opinion on the policy. Among these cities, the most interesting cases included Stockholm (with the aim to confirm or reject the existing charging scheme after a trial period), Milan (see below; Beria & Boggio, 2012), Edinburgh and Manchester (which both rejected the charging before implementation; Rye, Gaunt, & Ison, 2008; de Palma & Lindsey, 2011), and Gothenburg (which voted "no" after a short trial period).

Finally, numerous survey-based studies aimed at testing the attitude of populations toward urban road pricing (see Li & Hensher, 2012, for a general review and May et al., 2010; Jou et al., 2010; Dieplinger & Fürst, 2014; Pronello & Rappazzo, 2014, for specific cases).

Road pricing is not applied only at the urban scale. Not mentioning the numerous highway concessions existing around the world (Beria, Ramella, & Laurino, 2015), where tolls repay the construction costs, interesting cases of mileage-based pricing are represented by the recent German and Czech truck satellite-based, countrywide electronic tolling system (Arnold et al., 2010; de Palma & Lindsey, 2011). On the same idea, Al-Deek and Moradi (2015) estimated the impact of a mileage pricing scheme for Florida, substituting the existing per-gallon tax with a distance- and time-based toll, capable of reducing externalities and raising funds for infrastructure reinvestment.

3. Milan experiment: History and characteristics

In 2007 the city of Milan studied a measure to handle the increasingly worrisome air pollution problem, in particular particulate matter (PM10 and lower fractions). Milan's high concentrations of PM10 are sharpened, with respect to similar cities, by the particular geographical context (lack of wind). However, a crucial factor is also the large number of diesel-fueled cars. Moreover, the city administrators' actions were "under observation" by the European Commission, which threatened to impose a fine if the environmental problem were not adequately controlled.

3.1 *Former pollution charge measure (Ecopass)*

The measure, called *Ecopass*, started experimentally on 2 January 2008 and represented the first case in Italy of an internalizing road charge. It was explicitly aimed at introducing the "polluters pay" principle through a variable daily entrance toll to be paid by vehicles entering the central city Monday through Friday between 7:30 and 19:30. The toll, ranging from €2 to €10, differed according to the Euro emission class of the vehicle. Ecopass also banned vehicles older than Euro 0 and allowed vehicles with low-emission engines (natural gas, LPG, electric, hybrid) free entrance. The toll could be paid either in advance or within the day after entrance. Of course, as a secondary effect, Ecopass acted also as a rough congestion-charge toll. For a more complete discussion of the Ecopass scheme, including full rules and detailed charges, refer to Rotaris, Danielis, Marcucci, and Massiani (2010). A summary of Ecopass main facts is provided in Table 1.

The Area subject to control was 8 km^2, in the central sector of the town, delimited by the *Cerchia dei Bastioni*, corresponding to

[1] It is not the place here for a full discussion of road-pricing theory, first introduced by Vickrey (1963) and Kain (1972). For a complete theoretical introduction, see Rouwendal and Verhoef (2006).

[2] But May, Koh, Blackledge, and Fioretto (2010) report costs accounting for 48% of revenues. In any case, it is the largest share among charging schemes in use.

Table 1. Main characteristics of Ecopass (Rotaris et al., 2010) and Area C.

	Ecopass	Area C
Type of toll	Daily cordon toll, allowing unlimited crossings during the day of the entrance	
Toll classes	5 (0 to 10 €/day)	1 (always 5 €/day)
Toll payment	Single prepaid, multiple entries prepaid (40% to 50% discount), yearly pass for residents, postpaid (within the next day).	Single prepaid, multiple entries prepaid (no discount); single residents; single service vehicles; multiple entries prepaid.
Primary purpose	Pollution reduction	Congestion reduction
Technology used	Automatic number plate recognition (ANPR)	
Revenues	12.4 M€ in 2008, 8–9 M€ in 2011. No official data on fines collected, but likely to be at least of the same magnitude, or more.	20.3 M€/2012 excluding fines.

the 16th-century city walls. The Area covered just 4.5% of the Milan city Area (much less when considering the larger metropolitan Area, as shown in Fig. 1). It included approximately 78,000 inhabitants out of 1.4 million but hosted 37% of Milan's total number of employees (2001 census), with average densities during the working hours of 40,000 persons per square kilometer. From a commercial viewpoint, the Area encompassed the core of Milan, containing 23% of total commercial activities (Bedogni, Pulpito, & Tosi, 2011).

The effects of the policy were measured only ex post. Pollutant emission reduction was quite large but not revolutionary and declined over time thanks to fleet renewal, which increased the number of exempt vehicles. For example, total PM10 concentrations emitted inside the central Area decreased from −23% during the first year (Rotaris et al., 2010) to −15% by the end of the trial period (Bedogni et al., 2011). In the mid-term, however, the measure effectively stimulated the substitution of older, high-polluting vehicles for environmentally friendly vehicles: in 4 years, newer cars entering the Area increased by 478% and commercial vehicles increased by 1400%. Consequently, the indirect environmental benefit spread to the rest of the city.

The traffic inside the tolled Area initially decreased −16%,[3] but further benefits in terms of an increase in commercial speed, patronage of public transport, and road safety were measured (Bedogni et al., 2011).

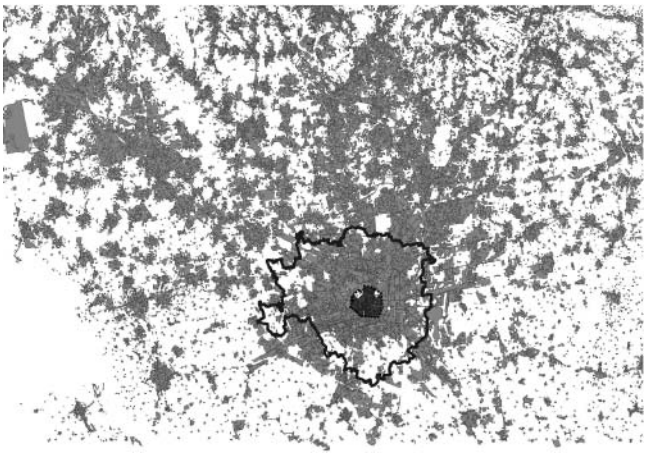


Figure 1. Milan metropolitan area, with city boundary and charged area. *Source.* Author's own representation.

The policy also revealed some limits:

1. A decline in the effectiveness as a congestion-control tool once the older vehicles were phased out. In 2008 approximately 20% of vehicles entering the cordon were tolled (Rotaris et al., 2010). In 2010 the share decreased to only 14.4%, and the rest were exempt (Commissione Ecopass, 2011).
2. Parking and localized congestions problems increased just outside the cordon.
3. A lack of clarity on the way in which revenues, initially €12.4 million per year (Rotaris et al., 2010) and later €8–9 million per year (Commissione Ecopass, 2011), were reinvested.[4]

3.2 *Referendum*

As its effectiveness in reducing congestion declined, Ecopass was reconsidered at the end of the trial period. The city's right-wing mayor, Letizia Moratti, established a technical-political commission to evaluate the measure. In the meantime, a group of local environmental associations promoted a referendum, which Moratti also accepted and promoted, on the topic. It took place on 12 June 2011. Quite surprisingly, before the referendum, nearly all political parties directly or indirectly supported, or at least did not explicitly oppose, extending the measure: governing right parties used the referendum to demonstrate their will to solve the pollution problem while left parties, more environmentally oriented, showed no opposition.

The main question of the referendum (too long to be fully reproduced here) asked the following:

> Do you wish for the Municipality of Milan to adopt and realize a plan of intervention to empower the public transport and the clean mobility alternative to cars, through the extension to all vehicles (excluding the zero-emissions ones) and the progressive broadening of the cordon charging, with the goal of halving traffic and pollution?

The question was followed by a list of specific measures (for example, doubling pedestrian Areas, extending the night metro opening, etc.) to be financed by the extension of the cordon and by parking pricing.

[3]That, however, must be depurated from exogenous effects, resulting in a net −6% to −7%.

[4]However, it must be noted that the total net resources invested in public transport is much larger, around €300 million per year.

As is evident, the referendum question was rather generic and blurred (who does not want a better environment?), asking citizens their opinion on an unspecified extension of the existing policy in exchange for funding a number of compensation measures, which may have influenced the vote.[5] In particular, the question failed to explicitly ask citizens about a so-called congestion charge. However, the ensuing debate made it clear to voters that the main point was a yes-or-no decision about the future of Ecopass, particularly its spatial and temporal extension.

The result was clear, with an overall *yes* rate of 79.12% and a good voter participation rate (usually very low) of 48%. Despite some differences, all city zones presented a positive vote, ranging from 71.7% to 89.2%.

This outcome can be partially explained by the perception of the pollution problem in the city and by communication campaigns. However, other elements were, in our opinion, crucial: the fact that only Milan residents voted (and not those living in the surrounding Areas), the extreme vagueness of the question, the compensation measures promised, and the presence of other referenda during the same round. For a full discussion of the referendum results, see Beria and Boggio (2012).

3.3 *New congestion charge (Area C)*

The referendum provided the necessary political legitimation to evolve the previous scheme toward a new, and possibly more effective, congestion-charge scheme. The new left-wing mayor, Giuliano Pisapia, started from the documents of the former commission (Commissione Ecopass, 2011) and issued an ex ante study analyzing possible policy extension scenarios (Bedogni et al., 2011). This document gave a base to all successive decisions and finally led to the definition of the new scheme and its implementation.

The new measure (see Table 1 for a comparison), now known as *Area C*, applies to the very same zone (Figure 2), remains a daily charge, and uses the same technological infrastructure as Ecopass.

The structure of the charge has changed. The remaining older vehicles (Euro 0 gasoline and Euro 0 to 3 diesel) may no longer circulate, while the others are admitted by paying a unique toll of €5 per day, regardless of the type of vehicle.[6] This change impacted differently the various types of cars and engines (Percoco, 2015): The newer ones previously not priced or priced 2€ saw a large decrease, while the older polluting engines, previously banned or priced 10€, did not change significantly but confirmed their downward trend.

Further discounts were introduced to the base 5€ toll to make it more acceptable to some categories. Commercial activities can be associated to service vehicles, which benefit from a reduced fare of €3 per day. Residents inside the cordon enjoy a package of 40 free entrances per year, and pay a discounted €2 per day for any extra entrance. All tolls can be prepaid or paid within a day after entering. A small fine of €15 has been introduced for delayed payments made within 7 days. Additionally, a cumulative ticket with underground parking was introduced. Automatic payment is allowed via Telepass radio technology (the same technology used to pay on Italian highways).

Unlike with Ecopass, detailed analyses of the expected impact of such charges were conducted and published before introducing the new measure. For example, it has been shown that with Ecopass, 50% of residents entered the cordon fewer than 40 times per year (Bedogni et al., 2011). Similar analyses have been published since implementation, confirming the expected impacts. We will further contribute to the topic in Section 5.

4. Effectiveness of Area C in 2012–2013

To fulfill their commitment to transparency, since the first months of 2012 the new city administration has periodically made available through various media reports and databases the Area C results. Data available are related to traffic (measured accesses: Nuccio & Tosi, 2012a) and pollution (simulated emissions: Bedogni, 2013). In addition, at the end of 2012, the administration issued a brief press release on revenue reinvestment, and similar documents are expected in the future.

In this section we refer to the main consolidated results for 2012 (with some elements for the first months of 2013), adding further comments. Unfortunately, to date, the only ex ante, ex post comparable data refer mainly to the total entrances and not to single users.

The ex post statistics available for entrances are detailed into five vehicle categories, to which different rules and prices apply. This allows more detailed analyses of users behavior during the period of application (see also Section 5). The groups are as follows:

i. Authorized: to this category belong a heterogeneous range of vehicles. They are taxis, buses, police, and other public vehicles, plus disabled people's and car-sharing cars. They are all preauthorized and recognized by the system, and are not charged to enter the Area.
ii. Low-emission vehicles: until 2015 low-polluting vehicles were allowed to enter free. These vehicles do not need to be preregistered, as they are automatically recognized by the system via connection with the national vehicle registry.
iii. Residents: cars belonging to citizens living in the priced Area can be preregistered to enjoy the 40 free entrances and a discounted charge of 2€ for the rest. The registration is not automatic nor compulsory, but obviously the majority of them took the opportunity. Residents owning low-emission vehicles do not need to register (until now) because those are free.
iv. Service vehicles: vehicles associated with commercial activities located in the priced Area. Like for residents, businesses must register their vehicle to obtain a discounted fare.
v. Others: the larger group includes all vehicles not belonging to the previous ones. They are ether the citizens living in other parts of the city or any other driver. The

[5] It is worth mentioning that, unlike Stockholm, where an information campaign on charging effectiveness took place (Hårsman & Quigley, 2010), in Milan the level of information to citizens, journalists, and scholars was definitely less satisfying. No detailed results were publicly available before the referendum but only after, thanks to the commitment of the new mayor.

[6] Vehicles longer than 7.5 m are banned.

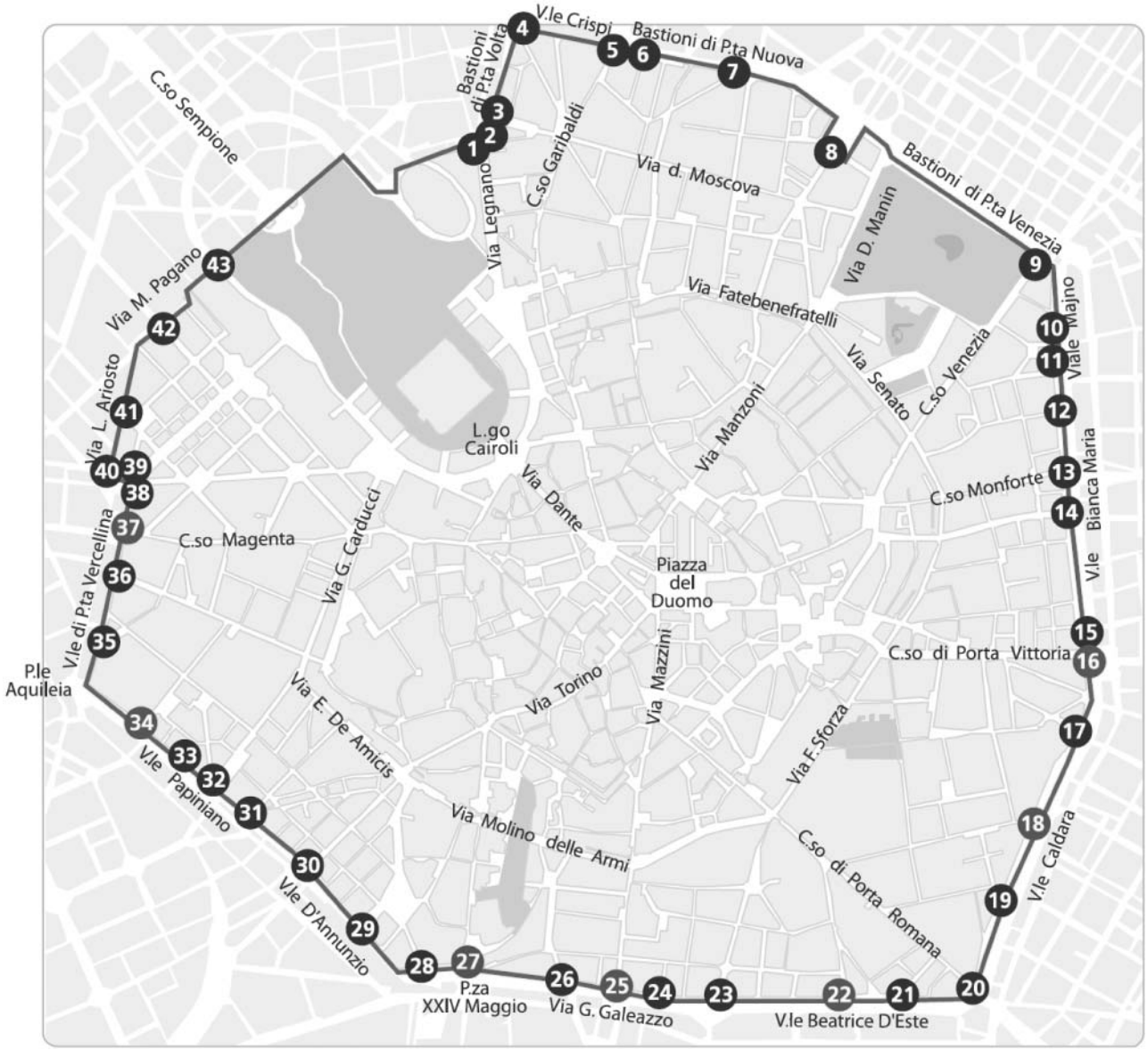


Figure 2. Milan cordon and the entrance points; light gray indicates those dedicated to public transport. *Source*. www.Area-C.it.

normal charge of 5€ is applied. The database of observations includes them from the first time they cross the cordon.

Of course, any vehicle that has never entered the Area (except those of preregistered residents) does not appear in any group, whatever its characteristics.

4.1 *Analysis of accesses*

Databases allow for analyzing 2012 access trends according to the different classes and groups of vehicles. In 2012 the measure was applied for 196 days, excluding Saturdays, Sundays, the days before Christmas, and between 1 and 16 January. In addition, starting in July, the state council enacted a 39-day forced suspension.

During the entire year (Figure 3), paying vehicles comprised approximately 59% of total admittances. Of these, the majority belonged to the *others* category. Authorized exempt vehicles accounted for 29% and low-emission exempt vehicles accounted for 12%. The majority of vehicles (85%) were passenger cars.

Apart from the total, it is interesting to analyze how the different groups of users behaved during the first year of application, as shown in Figure 4.

With respect to January, we can see a slight decrease for the *others* group (the majority of vehicles) and for low-emission vehicles. This decrease resulted from the fact that the first days of application were free in order to familiarize people with the new policy. Starting in February, the trend became rather constant until July. September showed slightly higher values, which remained constant until the end of the year. In general, the vast majority of normal users did not change their behavior during the first year of charge. Authorized vehicles showed a continuous trend because they were already preregistered (no learning phase).

It is more interesting to analyze the other two paying categories. Service vehicles constantly increased, finally doubling at the end of the year with respect to January. Our

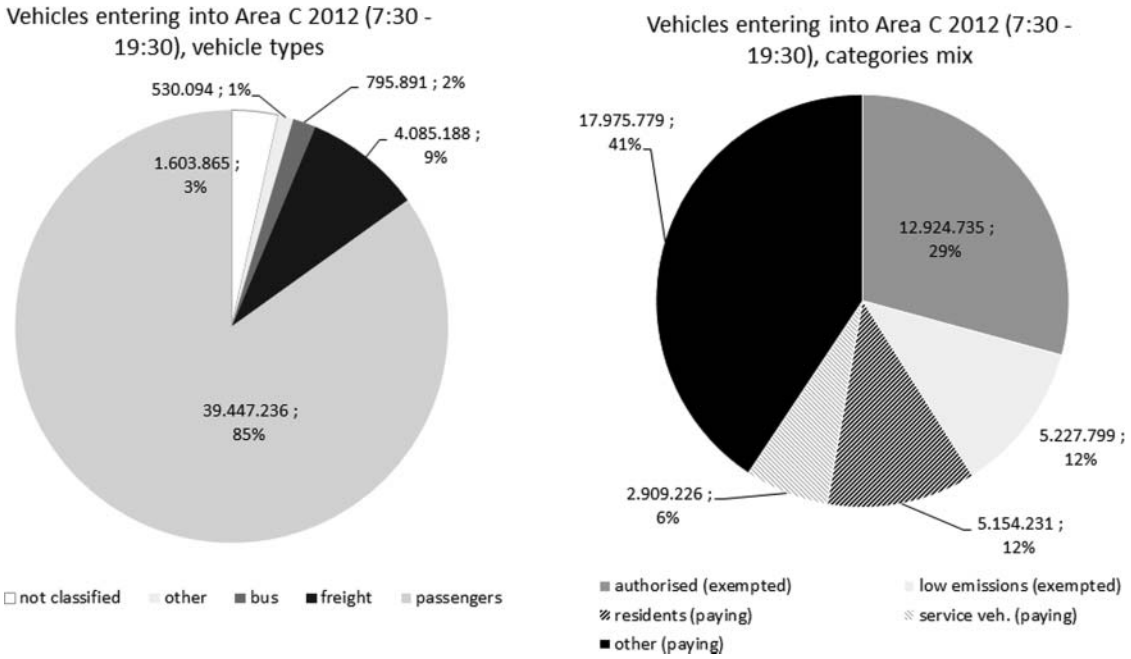


Figure 3. Total accesses during 2012, classified for vehicle types (left) or for user categories (right). *Source.* Author's own calculations on data from Nuccio & Tosi, 2012a.

interpretation is threefold. First, the commercial activities progressively registered their vehicles into the service-vehicle category to obtain the discount and consequently left the *others* group. Second, we hypothesize that commercial activities realized that the toll's impact on their business was marginal (as we will show). Last, but not least, they might have increased their vehicle use because congestion reduction benefits counterbalanced the toll.

Finally, *residents* also showed an increasing trend. Some were probably listed later in the group, but the postholiday increase can be interpreted differently: the numerous residents who seldom used their cars realized that the 40 free entrances were fully sufficient for their business and relaxed, finally using the car slightly more. The sharp decrease in July—much sharper than for other groups—is of some interest. A possible interpretation is that the number of residents in the central city Area decreased more than the city average during the summer holidays.[7]

For *residents* only, we have some figures also for the before the Area C period (Bedogni et al., 2011) because they had already registered for the Ecopass. During 2011, residents entered the Area between 7:30 and 19:30 around 14,500 times (10,200 unique vehicles). During 2012, this figure dropped to 12–13,000 cordon crossings per day, but most likely the residents are the less impacted group.

4.2 *Traffic reduction*

Before Area C's implementation, a plate-monitoring system for the Ecopass was already active in the very same Area. For this reason we have the daily access total numbers both before and after the start of Area C. Performing a comparison, Nuccio and Tosi (2012a) estimated the pricing's effect (see Figure 5).

During the weeks of Area C's application, the number of vehicles that entered the cordon (7:30 to 19:30) between 2012 and 2011 fell by an average of 31.1%, the equivalent of 41,000 fewer vehicles every day. The reduction was, however, slightly differentiated across the 50 weeks of 2012. According to data depicted in Figure 5, in fact, the weeks before July showed an average reduction in traffic of about 33.6% while the last period of the year showed a reduction of 26.7%. Some peak weeks (for example, the 14th) presented irregular figures usually related to holidays or specific events in the city that shift a few days every year (typically Fashion Week and a big furniture fair during the spring). These periods can be clearly seen in Figure 6.

Looking at the comparison, we also have a strong counterproof that the reduction is related to the existence of pricing (also found by Percoco, 2013). As already mentioned, between 26 July and 16 September, the Council of State suspended Area C. Part of the suspension occurred in August (during holidays), but part occurred during the first 2 weeks of September, which correspond to full working days. During the suspension, the figures of 2012 go back to 2011 level, demonstrating that the exogenous variation of traffic (not due to Area C or become structural) was negligible. The same happened for the 50th week (Christmas), when Area C was also suspended to foster shopping activities.[8]

Measuring the effect on the rest of the city is more difficult. For the first months (until April 2012), a synthetic measure of overall city traffic was published (Nuccio & Tosi, 2012b). According to this measure, the total traffic in Milan decreased less than 7% between 2011 and 2012. However, the difference between priced hours and the rest of the day was extremely small (−7% vs.

[7] August, the typical holiday month, is not priced and is excluded from the chart; thus, it cannot support this hypothesis.

[8] The suspension did not include the banned vehicles.

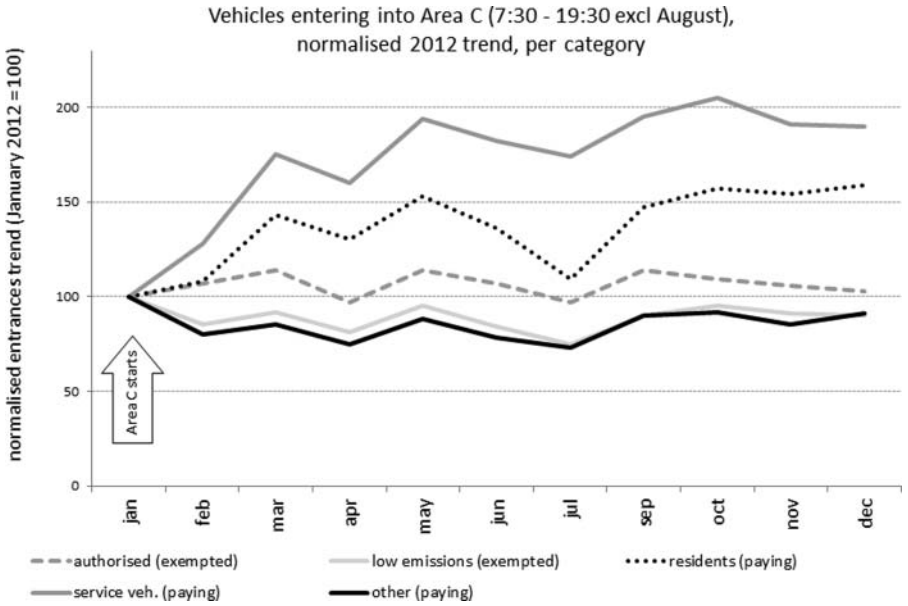


Figure 4. Analysis of trends of user groups entrances: normalized total accesses through the cordon during 2012, January 2012 = 100. *Source*. Author's own calculations on data from Nuccio & Tosi, 2012a.

−6.6%). Similarly, Mussone, Grant-Muller, and Laird (2015) found just a small increase in traffic speed for the city (+4.31 at 9:00), smaller than for the priced Area (+9.07 at 9:00), and assumed that it is mainly due to exogenous factors. This means that, on a city basis, the effect was negligible and the measured traffic decrease was mainly due to other factors or policies.

4.3 *Hourly accesses*

One effect of this kind of charging is to shift the accesses before and after the priced period, in this case before 7:30 and after 19:30. This may reduce the overall traffic reduction effect, but the shift, if the pricing period is correctly defined, should be toward uncongested hours. Figure 7 represents the average 2012 entrances in the cordon, with a segmentation every 30 min.

The shift is well visible, especially after 19:30, where a peak appears just after the end of the priced period. Another peak, of lower entity, is visible also before 7:30 in the morning. It means that numerous drivers, realistically the residents working outside and the leisure trips directed to restaurants and pubs in the center, wait until 19:30 before entering. The congestion effect of this counterflow of vehicles is, however, less problematic because the vehicles

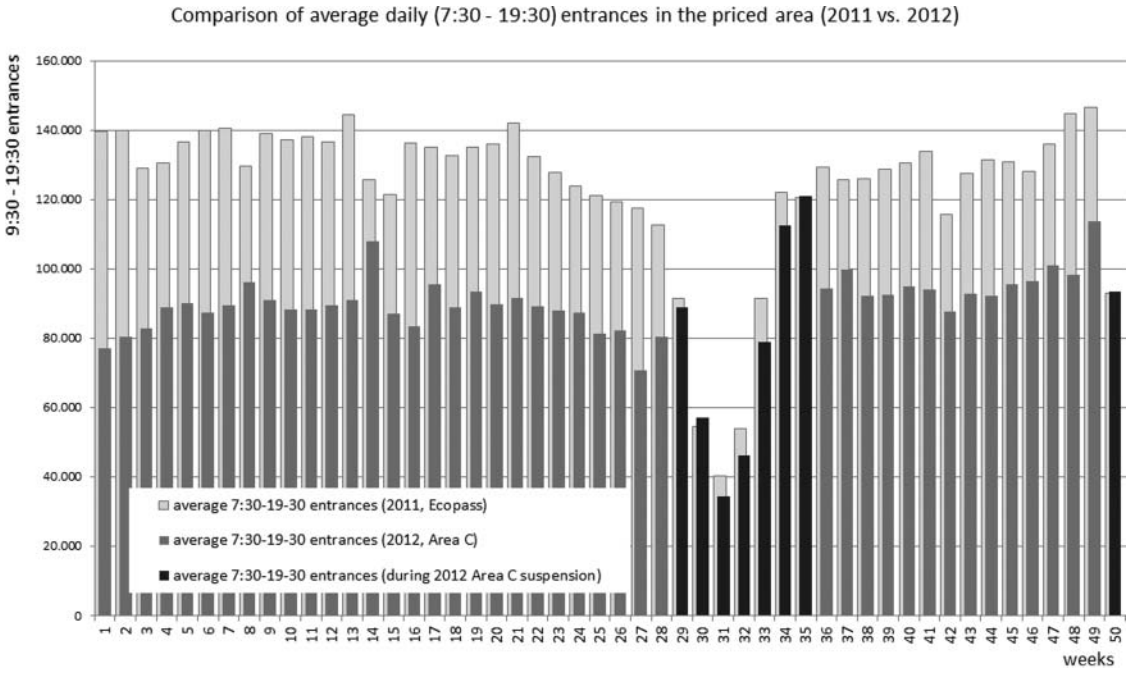


Figure 5. Comparison of 2011 vs. 2012 average daily entrances in the priced Area, during Area C priced hours (7:30–19:30). During weeks between 29 and 35 Area C was suspended. *Source*. Translation from Nuccio & Tosi, 2012a.

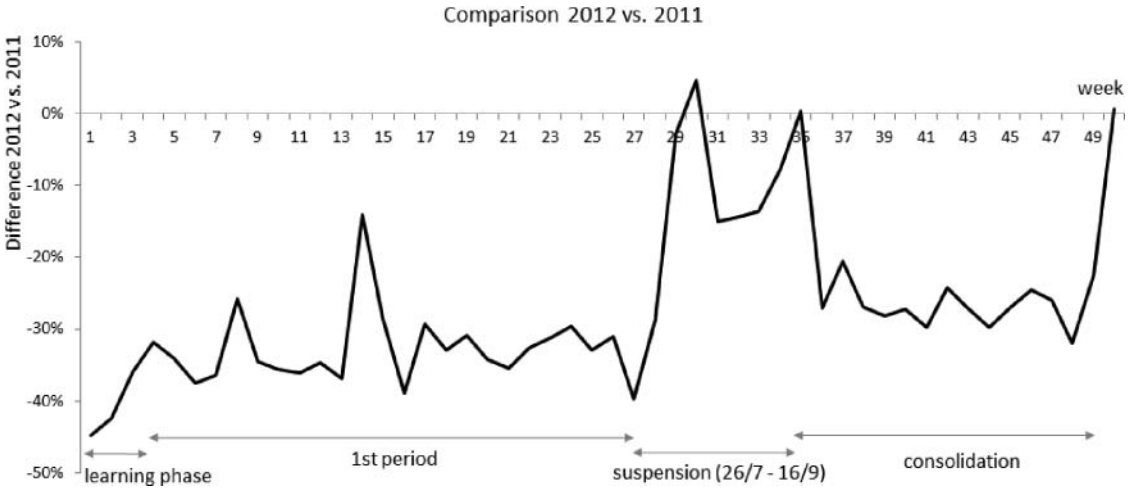


Figure 6. Analysis of access reduction (7:30–19:30), percentage of difference between 2012 and 2011.

are entering into an area that is becoming emptier as other people return home after work.

4.4 *Pollution reduction*

Bedogni (2013) analyzed Area C's effect on pollution reduction from traffic simulations, fleet composition, and unit emissions adapted for Milan from COPERT4 model. Simulations were run both for the day and for the priced time window and referred both to the priced Area and to the whole city. The results are not easy to interpret. In general, we see a sharp and continuous reduction in emissions in the past years, due both to overall traffic reduction and to fleet renewal. However, part of the reduction of the main pollutants can be attributed to Area C. For example, between 2011 and 2012 overall PM10 levels in the entire city and for the whole day fell 8%, while they fell 18% in the priced Area and during the activation of Area C. Table 2 summarizes the simulated effect of Area C at both scales.

The differences in terms of emissions reduction depend from the source of pollution. For example, the non-methane volatile organic compounds (NMVOC) are emitted mainly by motorcycles, which are not limited by Area C. Consequently, Area C's effect on that pollutant was smaller. Carbon dioxide (CO_2) can be used as a proxy for fuel consumption. The 10% reduction for the entire city was partly due to the overall traffic reduction and limiting the spread effect of Area C. Inside the Area and during pricing, instead, the reduction is higher, but an exogenous decrease of traveled kilometers exists anyway.

Percoco (2013) analyzed empirically the concentration of pollutants in Milan before and after Area C introduction. He found a positive effect, but limited to the short run. The analysis of the discontinuity due to the Council of State stop during summer confirmed that, as soon as the toll ended, the shift toward less-polluting vehicles returned to the previous situation.

4.5 *Revenues and reinvestment*

One of the main criticisms of the former Ecopass pricing regarded the lack of transparency (see Section 3.1) regarding the use of collected revenues. The new city administration

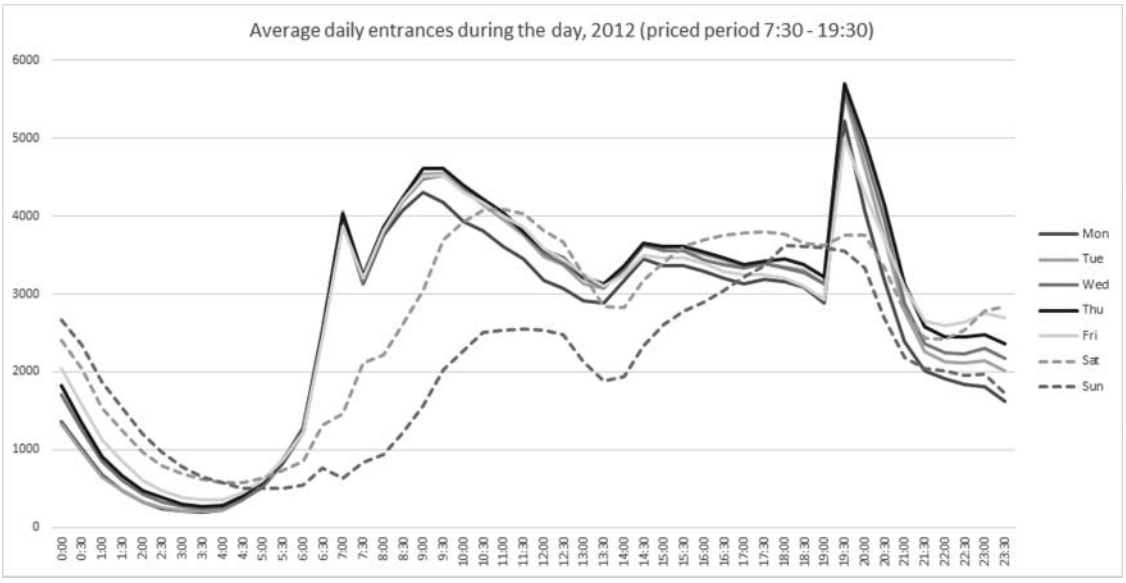


Figure 7. Trend of entrances during the day divided per day of the week. Average values: The priced period is between 7:30 and 19:30.

Table 2. Reduction of emitted pollutants between 2011 and 2012, simulation.

	Time period	
	0:00–24:00 (whole city)	7:30–19:30 (priced area)
PM10 total	−8%	−18%
NH_3	−20%	−42%
NO_x	−4%	−18%
NO_2	−8%	−25%
NMVOC	−4%	−14%
CO_2	−10%	−35%

Source. Bedogni, 2013.

promised more transparency on all aspects concerning Area C. Consequently, we know how much money was collected during the first 12 months and how it was reinvested (Table 3).

The running costs of the system were relatively low compared with other cities because the investment cost was already sunk: The control cameras were financed in the past for access control and had previously been used for Ecopass.

Also, given the importance of the redistribution (Ison & Rye, 2005; Marcucci & Marini, 2003), the reinvestment was totally concentrated on public transport and sustainable mobility policies. In particular, for 2012, €10 million was spent to finance a public transport supply increase and €3 million was spent to install 46 new bicycle-sharing stations.

5. Group behavior

In addition to quantifying traffic-reduction effects, policy makers should show interest in understanding who is more or less impacted by a policy (Eliasson & Mattsson, 2006; Karlström & Franklin, 2009). This is also related to the acceptability of road-charge measures, as some groups are more vocal while others are more likely to accept pricing even if heavily impacted.

A recent survey (Beria, Mariotti, Maltese, & Boscacci, 2013) based on 1,189 interviews shows that just a small share of drivers declared they were significantly impacted by Area C when making overall travel decisions. In detail, 56% of respondents stated that Area C did not negatively affect, or only marginally affected, their trips. Nineteen percent affirmed that they reduced car use, but only partially because of Area C. Just 19% decided to cease some trips (not necessarily all) due to the charge and 6% of those surveyed preferred to pay the toll.

Even if available data is not yet sufficient to understand the behavior of single users before and after the implementation, we can at least analyze the frequency of access during 2012 for the different groups and, eventually, the economic impact upon them. Published data (Nuccio & Tosi, 2012a) based on individual plate numbers of the vehicles recorded by the Area C scheme tell us the number of times they entered in a year.[9] Figure 8 represents all 196 days of active Area C during 2012 with the cumulated frequency of access for the five groups. The percentage tells how many crossers entered up to the corresponding number of days. The more a curve moves to the lower right side, the more the group is made of systematic crossers, because higher shares of them enter many times.

Table 3. Area C revenues 2012 and reinvestment.

		excluding fines
Revenues	20,300,000 €	(approx. 17 M€/year)
Running costs	7,100,000 €	
PT increase	∼10,000,000 €	+3370 km/day tram and bus, +4900 km/day metro
New bike-sharing stations	∼3,000,000 €	+ 46 stations

Source. Press release 15 January 2013.

The groups exhibit substantial differences. The large majority of paying vehicles that belong to the *others* group was clearly that of occasional crossers: For example, 48% entered the Area just once in a year and 90% entered no more than 12 times in a year. Only 1,800 cars entered more than 150 times, that is, the 0.22% of the *others* or 0.17% of all vehicles. On the other side, quite obviously, the *residents* and the *authorised* groups were the most systematic. Of the registered residents, 7.5% never crossed the cordon in a year[10] and another 2.5% entered just once, but half entered 20 times or more. The *authorized* category includes buses and other public vehicles, which intrinsically have very high entrance rates.

Quite interestingly, we can compare the behavior of the *residents* group before and after Area C, because residents were already subject to register their car before Area C to cross Ecopass cordon (Bedogni, 2011). Figure 9 shows how, as expected, residents become less systematic in 2012 with respect to 2011. For example, while in 2011 around 70% of vehicles entered no more than 60 times, in 2012 they increase to around 80% of the total. This means that the high-frequency drivers limited the number of crosses per year. Not surprisingly, the first part of the curve does not change: *residents* have 40 free entrances, so the *occasional crossers* do not need to decrease their car use.

The sole number of entrances reveals little about the real monetary impact of drivers' behaviors. To show that, we calculate the frequency of the annual amount paid by every group, by multiplying the number of entrances times the charge. This takes into account also the different pricing applied to the groups (free for authorized and low-emission vehicles, €3 for service vehicles, €5 for other vehicles, and €2 for residents after the 40 free entrances).

Not surprisingly, Figure 10 shows that *service vehicles* felt the highest economic impact, except a very small group of frequent crossers belonging to the *others* group. However, the amounts paid yearly by single vehicles were always quite limited, as occasional crossers largely prevailed in all groups. Among those entering at least once in 2012, 96% of residents, 97% of other users, and 87% of service vehicles paid less than €1 per day, on average.

Among *service vehicles*, which might present more rigidity in demand than private cars, occasional trips prevailed, and consequently annual tolls remained low: 57% paid less than €50 and 82% paid less than €150. Only 476 vehicles (2% of the total) paid more than €462 and just two of them entered every day, paying €588 during the entire year of 2012.

[9] The system is on also during night, but the analyses provided refer to the charged period only.

[10] Residents can preregister their vehicle, so we also know how many of them never cross.

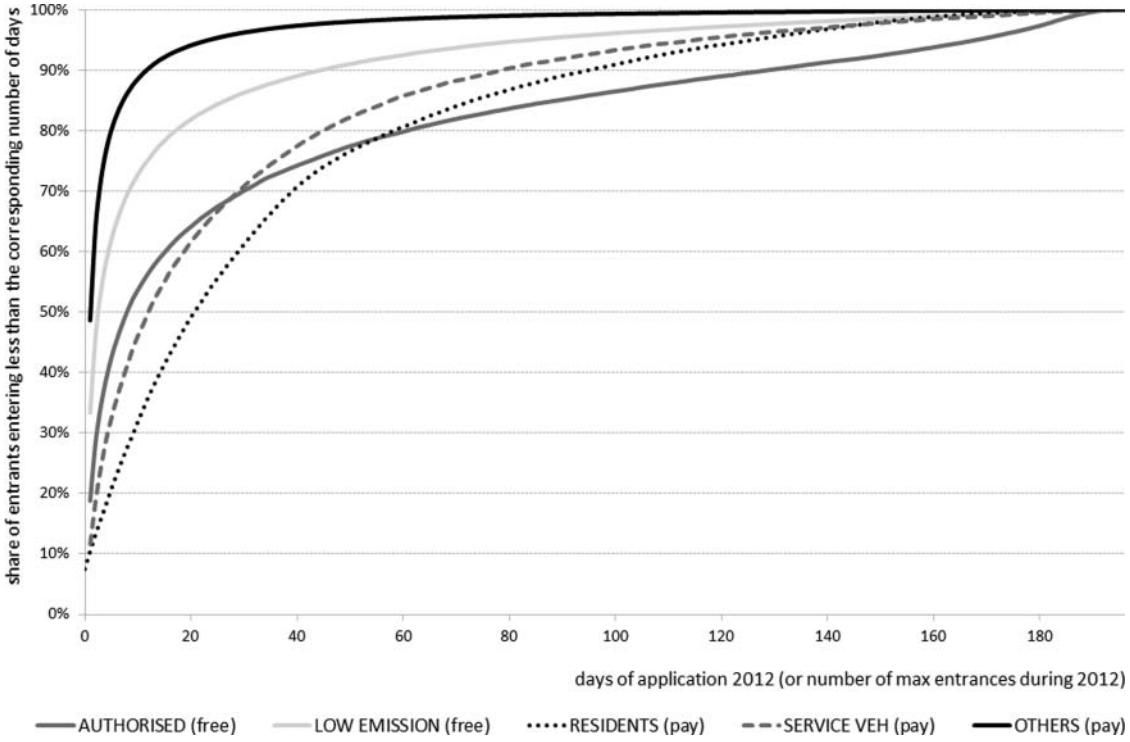


Figure 8. Cumulative share of yearly entrances for the 196 days of application of Area C during 2012, per user group.

Nonregistered cars (*others*) show the most concave curve; that is, very few were frequent crossers, and many were occasional crossers. Eighty-eight percent paid less than €50 and 96% paid less than €150. A total of 16,187 vehicles (2% of the total) paid more than €255 and just seven of them entered every day, paying €980 during the entire year of 2012.

The majority of the *residents* group, amounting to 35,000 cars, felt no financial impact from Area C: only 10,000 paid the toll during 2012 at least once. The remaining, more than 70%, paid nothing because they did not use their car (7.5%) or because they used their car fewer than 40 times (63%), as they probably did also before (see again Figure 9). Among payers,

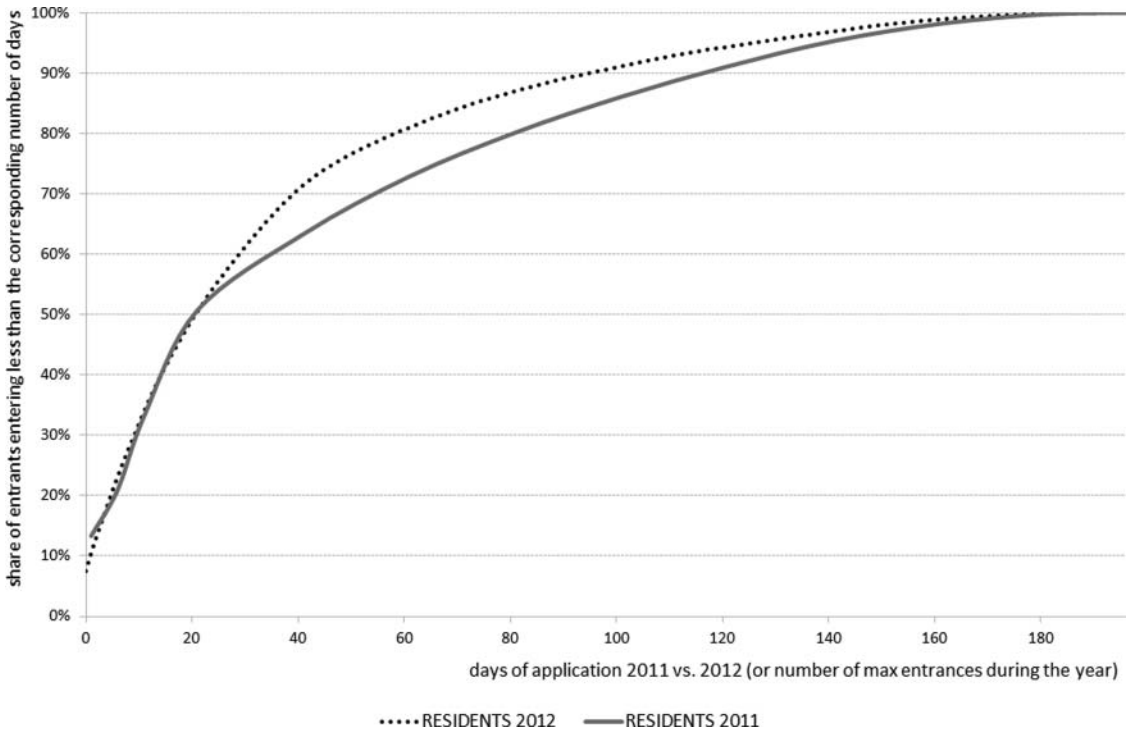


Figure 9. Comparison of 2011 vs. 2012 resident group behavior: Cumulative share of yearly entrances.

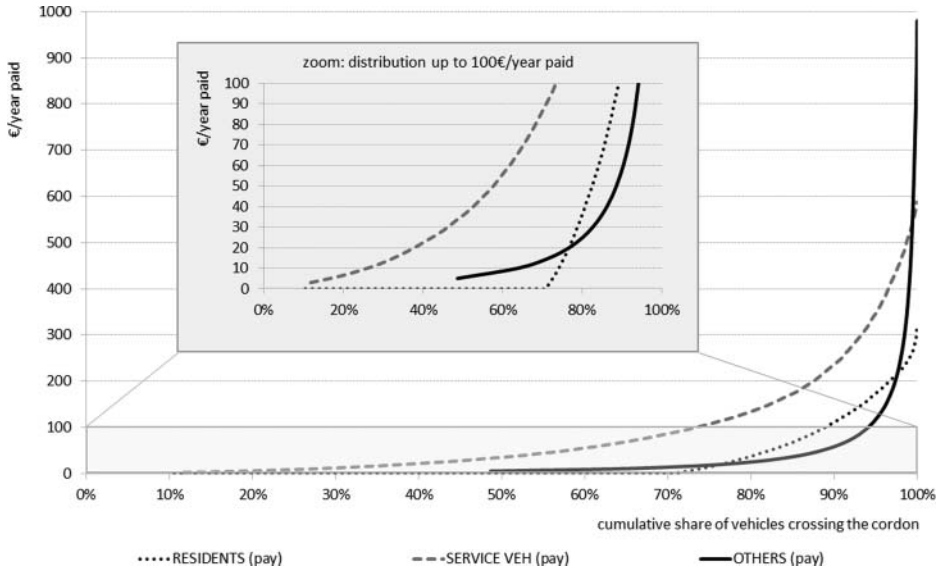


Figure 10. Distribution of yearly tolls paid during 2012, per user group.

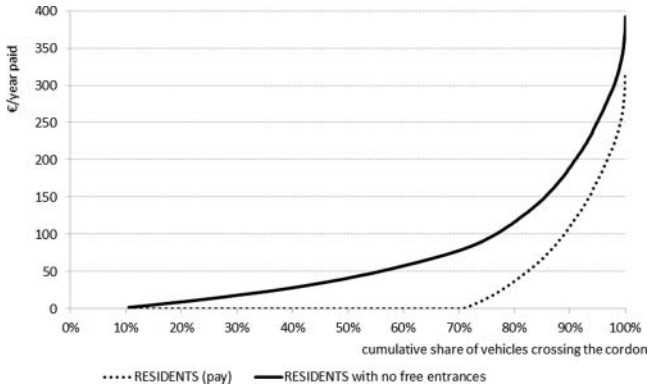


Figure 11. Comparison of residents' distribution of tolls paid during 2012, with and without the 40 free entrances (assuming rigid demand).

just 700 vehicles (2% of the total) paid more than €224 and just one single car entered every day, paying €312 during the entire year of 2012.

These figures explain quite clearly why the measure was accepted by citizens despite its relevant effects in terms of reduced traffic: Among the citizens (and residents in particular), an absolute minority felt the impact of an amount of €200 to €300 per year. Nearly all of Milan drivers paid no more than a few euros per year,[11] as they entered the Area just occasionally. In addition, many of the frequent enterers might not be Milan residents or be external suppliers and thus were not vocal against the pricing. The impact on residents was substantially reduced by the 40 free entrances policy. Figure 11 shows what would have been the impact without the free entrances, hypothetically assuming rigidity of demand. The majority of charges is paid by the many occasional users. Very few were instead the systematic users, who enter and pay large amounts in a year.

Of course all drivers interested potentially felt an impact, forced to choose between changing their trips or paying the fee, but the overall monetary impact on the average citizen and city user seems limited and did not cause protests. Additionally, already before the implementation of Area C just 19.5% of people entered the central city by car; the rest used public transport, rode bicycles, or walked.[12] The rest of the city (and its drivers) remain unaffected by the toll except for some border effect (not measured) and a very limited reduction of congestion.

6. Conclusions

This article presented the history and the effects of the Milan congestion charge Area C, introduced in the city during 2012 in place of the former pollution charge, Ecopass, which had existed since 2008 in an experimental form. Both versions of the charge were applied to the same zone: The historical center of the city, delimited by the former 16th-century walls. The Area is very delicate, presenting both a complex historical urban structure and a very high density of mixed functions, mainly tertiary and commercial but also residential.

Some years after its introduction, Ecopass's traffic reduction effect progressively declined due to the substitution of older vehicles with newer and cleaner ones exempt from payment. For this reason, the city administration organised a referendum to check citizens' attitudes toward an extension of the pricing. The referendum result was positive (79% voted yes) and the new administration enacted the new toll. The new structure requires, between 7:30 and 19:30, a flat €5 per day toll for nonregistered vehicles, €2 per day for residents (above 40 free entrances), and €3 per day for vehicles associated to commercial activities.

This article aimed to draw a picture of Area C's impacts and effects 1 year after its introduction. In particular, we showed the

[11] Of course, we are not discussing the impact on the generalized cost for those renouncing driving and shifting to public transport, bicycles, other destinations, or even cancelling their trips.

[12] Despite the very low share of private transport in the Area, it suffered from some congestion due to its urban structure (basically that of a medieval city) and due to its high density of activities, especially tertiary and commercial.

Table 4. Summary of Area C results, one year after introduction, with respect to 2011.

	Whole city	Priced area
Traffic reduction 7:30 – 19:30 Jan-Jul		−33.6%
Traffic reduction 7:30 – 19:30 Sep-Dec		−26.7%
Traffic reduction 7:30 – 19:30 whole year	∼−0,4%	−31.1%
PM reduction	−8%	−18%
CO^2 reduction	−10%	−35%

effectiveness of Area C in terms of traffic reduction, pollution reduction, and total revenues. We also discussed the behavior of user groups after implementation. The traffic reduction in 2012 (Table 4), with respect to the previous 2011 pricing scheme, was 31.1% inside the priced Area. However, two peaks of entrances appear just before and after the priced period, particularly at 19:30. As a further positive side effect of the pricing, particulate matter and CO_2 emissions in the Area fell by 18% and 35%, respectively. The policy generated a net revenue after running costs of approximately €13 million/year (Table 5), fully reinvested by the city administration in extra public transport supply and in the extension of bicycle sharing. Further revenues come from fines, but the precise amount is not known.

In addition, we analyzed the distribution of charges on user groups, quantifying the impact of the toll in terms of the average yearly amount paid. The analyses of individual frequency of entrances tell us that the large majority of vehicles crossing the cordon in 2012 were occasional. The groups of *service vehicles* and *residents* showed less occasional behavior. The pricing policy adopted (discounts and free entrances) substantially decreased the toll burden and facilitated the toll's acceptance of most affected groups. No more than a small percentage of users paid more than €1 per day on average (4% of residents, 3% of other users, and 13% of service vehicles). At the opposite end of the spectrum, the most affected users included just seven individuals (among some millions of different vehicles), who entered daily and paid €980 during the entire year of 2012.

These figures, together with a generalized perception of the pollution and congestion problems, explain the overall nonnegative attitude of the city toward road pricing, with the unavoidable exception of a few stakeholders. The center experienced a sharp reduction in traffic, which benefited both residents and city users and facilitated the commercial speed of cars and public transport. The most affected categories were favored with discounted tolls, thereby reducing the opposition but slightly increasing the number of total entrances. The numerous occasional drivers paid a negligible monetary burden on a yearly basis. In addition, revenues were reinvested in sustainable transport measures. Many city users and residents felt no negative effects because they already used public transport to enter the Area or because they did not systematically use a car in the central city Area.

Table 5. Summary of Area C financial results, one year after introduction, with respect to 2011.

Revenues	Running cost	Reinvestment
20.3 M€/year	7.1 M€/year	∼13 M€/2012

The experience of Area C and of its referendum can provide useful lessons for other cities:

1. As seen in Stockholm (Henriksson, 2009; De Borger & Proost, 2012; Hensher & Li, 2013; Eliasson, 2014), voting on a referendum after an initial test phase helps make the acceptance of the toll higher. The main explanation is that citizens already know the costs (the toll) and the related benefits (the effect on congestion and liveability) and emotive arguments influence them less. A counterproof is provided by the Edinburgh referendum (Rye et al., 2008), which failed also because of the lack of a test phase.
2. Where, like in Milan, the use of a car in sensitive central Areas is mainly occasional (for example, for shopping or to access administrative offices) as commuters already use public transport, a road charge provides important traffic effects (−30% approximately in our case). This is true even if the average toll paid yearly by each car is very small and consequently does not represent an unacceptable burden for the citizens. We can say that the avoided trips are those associated with lower utility, such as those for leisure, or because better alternatives exist (public transport or bicycle).
3. The definition of appropriate discounts for the most affected users (because they are systematic), such as commercial vehicles or residents, helps in gaining their support: a relatively small yearly cost in exchange for a more liveable neighborhood and increased road space.
4. The effect in terms of reduced traffic is not yet structural, as the temporary stop occurred during summer 2012 clearly showed. More time is needed to reshape the travel needs, the land use, and the users' personal habits in a way that does not need an exogenous policy action to obtain the same travel choices.
5. Of course, many other practical elements contribute to the success of the measure and need some time to be fine-tuned: Enforcement must be effective, payment systems must be easy and flexible (both for occasional and systematic users), and border effects must be controlled.

In conclusion, the effects of Area C are measurable and generally positive, even if concentrated only to the priced Area and not yet structural. Moreover, they were in line with the initial expectations. The policy did not show substantial acceptance problems: The population, with the exception of some legitimate resistance from specific categories, accepted the measure because their resulting private costs were smaller than their overall perceived benefits.

Acknowledgment

The author thanks Luca Tosi from AMAT for comments and suggestions.

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